Lecture Scribe Guidelines

L6_S2_A.pdf
PDF

1. Purpose and Evaluation Context
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This task evaluates: Precision in prompt adherence; Faithfulness to provided material; Disciplined and reproducible use of AI.
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Reasoning must reflect how conclusions follow, not external intuition.

6. Structure Requirement
Organize the scribe using clear sectioning that mirrors the lecture structure (for example): Random Variables; Bernoulli Random Variable; Binomial Distribution; Geometric Distribution; Poisson Distribution.
Logical dependencies between concepts must be explicit.

7. Output Constraint
Output only the lecture scribe. No commentary before or after the scribe. No meta explanations about the process.

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9. Instruction to Begin
Generate the lecture scribe now, strictly following all constraints above.

CSE400 – Fundamentals of Probability in Computing

Lecture 6: Discrete Random Variables, Expectation and Problem Solving
Instructor: Dhaval Patel, PhD
Date: January 22, 2025
L6_S2_A

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1. Random Variables: Motivation and Concept

Definition

A **random variable** X on a sample space Ω is a function

$$X : \Omega \to \mathbb{R}$$

that assigns to each sample point $\omega \in \Omega$ a real number $X(\omega)$.
L6_S2_A

Discrete Random Variables

In this lecture, attention is restricted to **discrete random variables**, i.e., random variables that take values in a set that is **finite or countably infinite**.

Although X maps into $\mathbb{R}$, the actual set of values

$$\{X(\omega) : \omega \in \Omega\}$$

is a discrete subset of $\mathbb{R}$.
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Visualization

- Sample points in Ω are mapped by X to points on the real line.
- Multiple sample points may map to the same real value.
- The distribution of a discrete random variable can be visualized using a **bar diagram**, where:

- The x-axis represents the values of the random variable.
- The height of the bar at value a is $\Pr[X = a]$. L6_S2_A

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2. Discrete vs Continuous Random Variables

Discrete Random Variable

- Countable support
- Described by a **Probability Mass Function (PMF)**
- Probabilities are assigned to **single values**
- Each possible value has strictly positive probability L6_S2_A

Continuous Random Variable (contrast only)

- Uncountable support
- Described by a **Probability Density Function (PDF)**
- Probabilities are assigned to **intervals**
- Each individual value has probability zero L6_S2_A

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3. Example: Tossing Three Fair Coins

Experiment Description

Three fair coins are tossed.
Let Y denote the **number of heads** that appear.

Possible Values

$$Y \in \{0, 1, 2, 3\}$$

Computation of Probabilities

Each outcome has probability $(1/2)^3 = 1/8$.

- $\Pr[Y = 0] = \Pr[(t, t, t)] = \frac{1}{8}$
- $\Pr[Y = 1] = \Pr[(t, t, h), (t, h, t), (h, t, t)] = \frac{3}{8}$
- $\Pr[Y = 2] = \Pr[(t, h, h), (h, t, h), (h, h, t)] = \frac{3}{8}$
- $\Pr[Y = 3] = \Pr[(h, h, h)] = \frac{1}{8}$ L6_S2_A

Validation

Since Y must take one of the values $0, 1, 2, 3$,

$$\sum_{i=0}^{3} \Pr[Y = i] = 1$$

which confirms a legitimate probability mass function.

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4. Probability Mass Function (PMF)

Definition

A random variable that can take on at most a countable number of possible values is said to be **discrete**. Let X be a discrete random variable with range

$$R_X = \{x_1, x_2, x_3, \dots\} \quad \text{(finite or countably infinite)}.$$

The function

$$P_X(x_k) = \Pr[X = x_k], \quad k = 1, 2, 3, \dots$$

is called the **Probability Mass Function (PMF)** of X.

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Property

Since X must take exactly one of the values x_k,

$$\sum_{k=1}^{\infty} P_X(x_k) = 1.$$

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5. PMF Example with Parameter λ

Given

The PMF of a random variable X is

$$p(i) = c\frac{\lambda^i}{i!}, \quad i = 0, 1, 2, \dots$$

where $\lambda > 0$.

Step 1: Find c

Since probabilities must sum to 1,

$$\sum_{i=0}^{\infty} p(i) = 1$$

$$c\sum_{i=0}^{\infty}\frac{\lambda^i}{i!}=1$$

Using the identity

$$e^{\lambda}=\sum_{i=0}^{\infty}\frac{\lambda^i}{i!},$$

we obtain

$$ce^{\lambda}=1 \quad \Rightarrow \quad c=e^{-\lambda}.$$

Step 2: Compute $\Pr[X=0]$

$$\Pr[X=0]=e^{-\lambda}\frac{\lambda^0}{0!}=e^{-\lambda}.$$

Step 3: Compute $\Pr[X>2]$

$$\begin{aligned}\Pr[X>2]&=1-\Pr[X\le 2]\\&=1-\big(\Pr[X=0]+\Pr[X=1]+\Pr[X=2]\big)\\&=1-\left(e^{-\lambda}+\lambda e^{-\lambda}+\frac{\lambda^2}{2}e^{-\lambda}\right).\end{aligned}$$

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6. Bayes' Theorem (Recap)

Statement

Using

$$\Pr(A\cap B)=\Pr(B\mid A)\Pr(A),$$

we obtain **Bayes' Formula**:

$$\Pr(B_i\mid A)=\frac{\Pr(A\mid B_i)\Pr(B_i)}{\sum_j \Pr(A\mid B_j)\Pr(B_j)}.$$

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Terminology

- $\Pr(B_i)$: **a priori probability**
- $\Pr(B_i\mid A)$: **posterior probability**

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7. Bayes' Theorem Example: Auditorium with 30 Rows

Setup

- Auditorium has 30 rows.
- Row 1 has 11 seats, Row 2 has 12 seats, ..., Row 30 has 40 seats.
- A door prize is awarded by:
  1. Randomly selecting a row (each row equally likely),
  2. Randomly selecting a seat within that row (each seat equally likely).

Tasks

1. Compute $\Pr(\text{Seat } 15 \mid \text{Row } 20)$.
2. Compute $\Pr(\text{Row } 20 \mid \text{Seat } 15)$.

Results

- Since Row 20 has 30 seats:

$$\Pr(\text{Seat } 15 \mid \text{Row } 20) = \frac{1}{30}.$$

- Using Bayes' formula:

$$\Pr(\text{Row } 20 \mid \text{Seat } 15) = \frac{\Pr(\text{Seat } 15 \mid \text{Row } 20)\Pr(\text{Row } 20)}{\Pr(\text{Seat } 15)}.$$

From the lecture computation:

$$\Pr(\text{Seat } 15) \approx 0.0342, \quad \Pr(\text{Row } 20 \mid \text{Seat } 15) \approx 0.0325.$$

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Sources

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Lecture 6: Discrete Random Variables, Expectation and Problem Solving
Instructor: Dhaval Patel, PhD
Date: January 22, 2025

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1. Random Variables

Definition

A **random variable** X on a sample space Ω is a function

$$X : \Omega \to \mathbb{R}$$

that assigns to each sample point $\omega \in \Omega$ a real number $X(\omega)$.

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Discrete Random Variables

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Although X is defined as a mapping into $\mathbb{R}$, the actual set of values

$$\{X(\omega) : \omega \in \Omega\}$$

is a discrete subset of $\mathbb{R}$.

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Visualization of a Distribution

- The x-axis represents the values that the random variable can take.
- The height of the bar at a value a is the probability $\Pr[X = a]$.
- Each probability is computed as the probability of the corresponding event in the sample space.

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2. Discrete and Continuous Random Variables

Discrete Random Variable

- Countable support
- Probability Mass Function (PMF)
- Probabilities are assigned to single values
- Each possible value has strictly positive probability

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Continuous Random Variable

- Uncountable support
- Probability Density Function (PDF)
- Probabilities are assigned to intervals of values
- Each individual value has probability zero

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3. Example: Tossing Three Fair Coins

Experiment

Three fair coins are tossed.
Let Y denote the number of heads that appear.

Possible Values

$$Y \in \{0, 1, 2, 3\}$$

Computation of Probabilities

Each outcome of three coin tosses has probability $(1/2)^3 = 1/8$.

- $\Pr[Y = 0] = \Pr[(t, t, t)] = \frac{1}{8}$
- $\Pr[Y = 1] = \Pr[(t, t, h), (t, h, t), (h, t, t)] = \frac{3}{8}$
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Legitimacy of the PMF

Since Y must take one of the values $0, 1, 2, 3$,

$$\sum_{i=0}^{3} \Pr[Y = i] = 1.$$

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4. Probability Mass Function (PMF)

Definition

A random variable that can take on at most a countable number of possible values is said to be **discrete**.

Let X be a discrete random variable with range

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The function

$$P_X(x_k) = \Pr[X = x_k], \quad k = 1, 2, 3, \dots$$

is called the **Probability Mass Function (PMF)** of X. L6_S2_A

Property

Since X must take exactly one of the values in its range,

$$\sum_{k=1}^{\infty} P_X(x_k) = 1.$$

5. PMF Example with Parameter λ

Given

The PMF of a random variable X is

$$p(i) = c\frac{\lambda^i}{i!}, \quad i = 0, 1, 2, \ldots$$

where $\lambda > 0$.

Determination of the Constant c

Since probabilities must sum to 1,

$$\sum_{i=0}^{\infty} p(i) = 1.$$

Substituting the PMF,

$$c\sum_{i=0}^{\infty} \frac{\lambda^i}{i!} = 1.$$

Using the identity

$$e^{\lambda} = \sum_{i=0}^{\infty} \frac{\lambda^i}{i!},$$

we obtain

$$ce^{\lambda} = 1 \quad \Rightarrow \quad c = e^{-\lambda}.$$

Computation of Probabilities

$$\Pr[X = 0] = e^{-\lambda}\frac{\lambda^0}{0!} = e^{-\lambda}.$$

$$\Pr[X > 2] = 1 - \Pr[X \leq 2]$$

$$= 1 - \big(\Pr[X = 0] + \Pr[X = 1] + \Pr[X = 2]\big)$$

$$= 1 - \left(e^{-\lambda} + \lambda e^{-\lambda} + \frac{\lambda^2}{2}e^{-\lambda}\right).$$

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6. Bayes' Theorem

Statement

Using

$$\Pr(A \cap B) = \Pr(B \mid A)\Pr(A),$$

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Computations

Probability of Seat 15 given Row 20
Row 20 has 30 seats, hence

$$\Pr(\text{Seat } 15 \mid \text{Row } 20) = \frac{1}{30}.$$

Probability of Row 20 given Seat 15
Using Bayes' formula,

$$\Pr(\text{Row } 20 \mid \text{Seat } 15) = \frac{\Pr(\text{Seat } 15 \mid \text{Row } 20)\Pr(\text{Row } 20)}{\Pr(\text{Seat } 15)}.$$

From the lecture computation,

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is a discrete subset of $\mathbb{R}$.
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Distribution Visualization

The distribution of a discrete random variable can be visualized as a bar diagram:

- The x-axis represents the values that the random variable can take.
- The height of the bar at value a is the probability $\Pr[X = a]$.

- Each probability is computed by evaluating the probability of the corresponding event in the sample space.

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PMF Validation

Since Y must take exactly one of the values $0, 1, 2, 3$,

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is called the **Probability Mass Function (PMF)** of X.

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Fundamental Property

Since X must take exactly one value in its range,

$$\sum_{k=1}^{\infty} P_X(x_k) = 1.$$

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L6_S2_A

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Probability of Row 20 given Seat 15
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